Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/11/2005
PI Name:	Moldover, Michael R. Ph.D.		
Project Title:	Acoustic Study of Critical Phenomena in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	National Institute of Standards and Technology		
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City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8360	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/21/2004	**End Date:**	04/20/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hall, Nancy Rabel	**Contact Phone:**	216-433-5643
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Flight Program:			
Flight Assignment:	Note: grant renewed 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gillis, Keith A (NIST) Shinder, Iosif I (NIST)		
Grant/Contract No.:	NNC04IA20I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using a double-Helmholtz acoustic resonator, also known as a Greenspan viscometer, we will measure the speed of sound and acoustic dissipation in a sample of xenon within 6 mK of the critical temperature. The speed of sound data will yield accurate measurement of the constant-volume heat capacity, and the quality factor of the resonator will be used to determine the bulk viscosity. These measurements will be extended to resonators made of different materials to test our understanding of thermoacoustic boundary layers. These measurements will be extended to off-critical densities to be certain that we obtained the correct critical density and to search for the predicted divergence of the second derivative of the chemical potential with respect to temperature. If the divergence is found, it will show that fluids do not have the symmetry of the Ising model in the usual thermodynamic variables. The objective of this research is to measure the bulk viscosity of xenon near its liquid-vapor critical point using a novel acoustic resonator. The new measurements will be at frequencies that are a factor of 4000 lower than previous measurements, allowing us to extend the range of useful data a factor of 60 closer to the critical temperature. The bulk viscosity (or second viscosity) is the transport coefficient associated with dilational motion of a fluid. The bulk viscosity of a gas is often negligible in comparison to the shear viscosity. However, systems such as critical fluids or polyatomic gases may require many collisions before their internal degrees of freedom come into equilibrium with translational degrees of freedom, and the bulk viscosity can become much larger than the shear viscosity. Our measurements in microgravity will provide the most rigorous possible test of the theory for the bulk viscosity, i.e., the theory for the time-correlations of pairs of critical fluctuations. The quality-factor data will also test the theory of the thermal boundary layer in heretofore inaccessible ranges of frequency and fluid properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental; thus, its consequences are not immediate; however they are general. We now understand the bulk viscosity of xenon near its critical point. The bulk viscosity is the property that dominates the attenuation sound (and shock waves) in near critical fluids. We plan to extend this understanding to off-critical densities. The universality of critical phenomena quantitatively predicts the bulk viscosity of, for example, near-critical steam. Thus, we are contribution to the modeling and control of shock waves in turbines that use near-critical steam to generate electricity.
Task Progress:	We used thermal convection to stir xenon in acoustic resonators. The stirring enabled us to make speed -of-sound and attenuation measurements within 1 mK of the critical point. This is a factor of 60 better than any previous acoustic measurements. The results were used to determine the critical-point anomalies in the thermodynamic (speed-of-sound, constant-volume heat-capacity) and transport properties (thermal conductivity and bulk viscosity) of xenon more accurately than ever before. A manuscript has been prepared for Physical Review E and is in review. The title is: Bulk Viscosity of Stirred Xenon near the Critical Point The abstract is: Following Gillis, Shinder, and Moldover [Phys. Rev. E 70, 021201 (2004)], we deduce the thermophysical properties of near-critical xenon from measurements of the frequencies and half-widths of the acoustic resonances of xenon maintained at its critical density in centimeter-sized cavities. In the reduced temperature range $10^{-3} < T-Tc)/Tc < 10^{-6}$, we measured the resonance frequencies and quality factors of six modes spanning a factor of 27 in frequency. As Tc was approached, the frequencies decreased by a factor of 2.2 and the quality factors decreased by as much as a factor of 140. Remarkably, these results are predicted (within +/-2 % of the frequency and within a factor of 1.8 of the quality factor) by a model for the resonator and a model for the frequency-dependent bulk viscosity zeta(omega) that uses no empirically-determined parameters. In addition to describing the present low-frequency data (120 Hz to 7.5 kHz), the model for zeta(omega) is consistent with ultrasonic (0.4 MHz to 7 MHz) velocity and attenuation data from the literature. However, the model predicts a peak in the temperature-dependence of the dissipation in the boundary layer that we did not detect. This suggests that the model overestimates the role of the bulk viscosity on the thermal boundary layer. In this work, the acoustic cavities were heated from below to stir the xenon, thereby reducing the density stratification resulting from earth's gravity. The stirring reduced the apparent equilibration time from several hours to the order of one minute and also improved the effective temperature resolution from 60 mK to approximately 2 mK, which corresponds to $(T-Tc)/Tc = 7x10^{-6}$.
Bibliography Type:	Description: (Last Updated: 09/16/2010)
Articles in Peer-reviewed Journals	Gillis KA, Shinder II, Moldover MR. "Thermoacoustic boundary layers near the liquid-vapor critical point." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Aug;70(2 Pt 1):021201. PMID: 15447479 , Aug-2004